

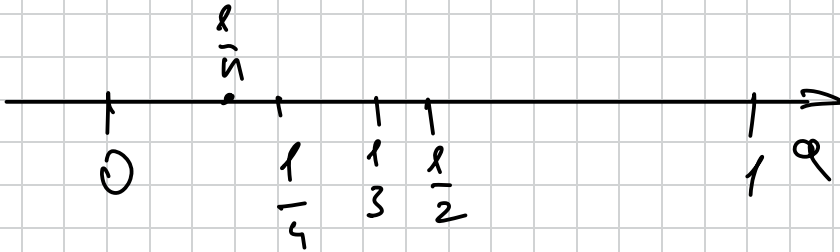


Физика

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Последовательности

$$a_n = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n} \right\}$$



a_n стремится к нулю — a_n есть предел
(сходится)

Предел — $\lim$

$$a_n \rightarrow 0$$

$$\lim_{n \rightarrow \infty} a_n = 0$$

$$b_n = 3 + \frac{2}{n}$$

$$\lim_{n \rightarrow \infty} b_n = 3$$

$$y = f(x)$$

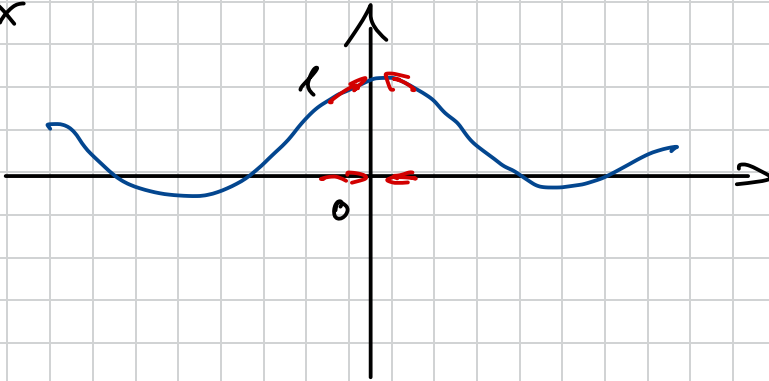
$$y = x^2$$

$$\lim_{x \rightarrow 2} x^2 = 4$$



$$y = \frac{\sin(x)}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$



$$v = \frac{\Delta S}{\Delta t} = \frac{S(t + \Delta t) - S(t)}{\Delta t}$$

$$S(t) = t^2 \quad v(3) = \rightarrow 6 \text{ m/c}$$

$$1. \Delta t = 1 \text{ c}$$

$$\Delta S = S(4) - S(3) = 16 - 9 = 7 \text{ m}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{7}{1} = 7 \text{ m/c}$$

$$2) 2. \Delta t = 0,1 \text{ c}$$

$$\Delta S = S(3,1) - S(3) = 0,61 \text{ м}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{0,61}{0,1} = 6,1 \text{ м/с}$$

$$3. 3. \Delta t = 0,01 \text{ c}$$

$$\Delta S = S(3,01) - S(3) = 0,0601 \text{ м}$$

$$v = \frac{\Delta S}{\Delta t} = \frac{0,0601}{0,01} = 6,01 \text{ м/с}$$

$$4. \Delta t = 0,001 \text{ c}$$

$$v = 6,001 \text{ м/с}$$

$v(3) \rightarrow 6 \text{ м/с}$ (мгновенная скорость)

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t}$$

$$\Delta S = (t + \Delta t)^2 - t^2 = 2t\Delta t + \Delta t^2 = \Delta t(2t + \Delta t)$$

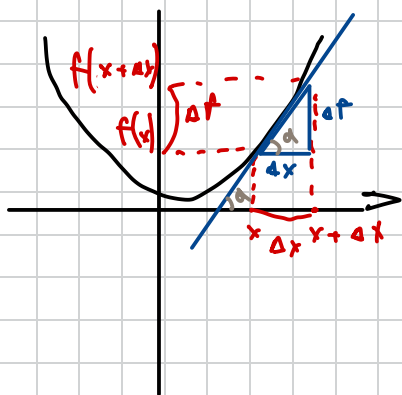
$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta t(2t + \Delta t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} 2t + \Delta t = 2t$$

$$v(t) = 2t$$

$$v(3) = 6$$

Производная

$$y = f(x)$$



$$\frac{\Delta f}{\Delta x} = \tan \alpha$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \tan \alpha$$

Производная

$$(x^a)' = a x^{a-1}$$

$$x^{-\frac{1}{2}} = x^{\frac{1}{2}} = \frac{1}{\sqrt{x}}$$

$$x^{\frac{1}{2}} = \sqrt{x}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

$$x^{\frac{2}{3}} = \sqrt[3]{x^2}$$

$$x^{\frac{5}{3}} = \sqrt[3]{x^5}$$

Таблица производных

$F(x)$	$f'(x)$
c	0
x	1
x^2	$2x$
x^3	$3x^2$
x^a	ax^{a-1}
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\operatorname{tg} x$	$\frac{1}{\cos^2 x}$

$$v = s'(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt} = \dot{s}(t)$$

d - символ, обозначающий дифференциал

$$a = \frac{\Delta v}{\Delta t} \longrightarrow \frac{dv}{dt} = \dot{v}(t) = \ddot{s}(t) = s''(t)$$

Правила дифференцирования

$$c = \text{const}$$

$$v = f(x)$$

$$1. (c \cdot v)' = c \cdot v'$$

$$(3x)' = 3 \cdot x' = 3$$

$$(2 \sin x)' = 2 \cos x$$

$$2. (v \pm v)' = v' \pm v'$$

$$(2x^2 - 3x)' = (2x^2)' + (-3x)' = 4x - 3$$

$$\left(\sin x - \frac{x^2}{2} \right)' = \cos x - x$$

$$3. (v \cdot v)' = v'v + vv'$$

$$(x \cdot \cos x)' =$$

$$= x' \cdot \cos x + x \cdot (\cos x)' =$$

$$= \cos x - x \sin x$$

$$(\pi \cdot x)' = x' \pi + x \pi'$$

$$= 0 + x \cdot 0 = 0$$

первое правило

$$4. \left(\frac{u}{v} \right)' = \frac{u'v - uv'}{v^2} \quad \frac{(2\sin x)' \cdot (x-1) - 2\sin x \cdot (x-1)'}{(x-1)^2} =$$

$$= \frac{2\cos x(x-1) - 2\sin x}{(x-1)^2}$$

$$(\tan x)' = \left(\frac{\sin x}{\cos x} \right)'$$

$$= \frac{\cos x \cdot \cos x - \sin x(-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x - \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$5. (u(v))' = u'(v) \cdot v'$$

$$\underbrace{(\sin 2x)}_u = \underbrace{\cos(2x)}_{v'} \cdot 2$$

$$= 2\cos 2x$$

$$\underbrace{(\cos x)^2}_v' = 2(\dots)' \cdot (-\sin x)$$

$$= -2\cos x \cdot \sin x$$

$$= -\sin 2x$$

Рассмотрим задачу:

$\xrightarrow{\text{дан}}$

$$m = 3 \text{ кг}$$

$$x(t) = 2 - 8t + t^2$$

$$x(t) = x_0 + 2t - \frac{at^2}{2}$$

$$1. v(t) = ?$$

$$2. v(5) = ?$$

$$3. t_{\text{остановки}} = ?$$

$$4. a(t) = ?$$

$$5. F(t) = ?$$

$$6. p(t) = ?$$

$$1. v = \dot{x}(t) = (2 - 8t + t^2)'$$

$$= -8 + 2t \quad v(t) = v_0 + at$$

$$2. v(5) = -8 + 10 = 2 \text{ м/с}$$

$$3. 0 = -8 + 2t_{\text{ост}} \Rightarrow t_{\text{ост}} = 4$$

$$4. a(t) = \dot{v}(t) = (-8 + 2t)'$$

$$= 2 \text{ м/с}^2$$

$$6. p = m v = 3 \cdot (-8 + 2t) =$$

$$= -24 + 6t$$

$$5.1 \quad F = m a = 3 \cdot 2 = 6 \text{ Н}$$

$$5.2 \quad F = m \frac{dv}{dt}$$

$$F = \frac{d(mv)}{dt} = \frac{dp}{dt}$$

$$= -24 + 6t = 6 \text{ Н}$$

$\triangleleft$ aus

$$x(t) = R \cos(\omega t)$$

$$y(t) = R \sin(\omega t)$$

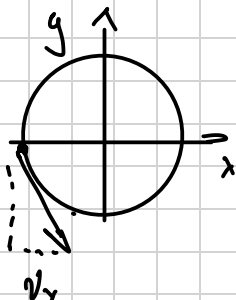
$$\omega = \text{const}$$

$$t = \text{const}$$

$$1. \quad x^2 + y^2 = R^2 \cos^2 \omega t + R^2 \sin^2 \omega t = R^2$$

$$x_0 = R \cos 0 = R$$

$$y_0 = R \sin 0 = 0$$



1. $y(x) - ?$

2. $\left| \vec{v}(t) \right| - ?$

3. $a_n(t) - ?$

$$2. \quad v_x(t) = (R \cos(\omega t))'$$

$$= R (-\sin(\omega t)) \cdot \omega$$

$$= -\omega R \sin(\omega t)$$

$$v_y(t) = (R \sin(\omega t))'$$

$$= R (\cos(\omega t)) \cdot \omega$$

$$= \omega R \cos \omega t$$

$$v(t) = \sqrt{v_x^2 + v_y^2} = \sqrt{\omega^2 R^2} =$$

$$= \omega R$$

